Problem with Distribution ?

: In many Scenario, the number of outcomes can be much larger & hence a table would be tedious to write down.

Worse still, the no. of possible outcome could be infinite, in which case, good luck writing a table.

Solution ? :⇒

What if we use a mathematical function to model the relationship between outcome and probability, which help to show the distribution graphically.

exp = throwing 3 dice together,

— x —

$\boxed{\text{Probability Distribution Function - PDF}}$

A probability distribution function is a mathematical function that describe the probability of obtaining different value of Random Variable in a particular probability distribution.

$$y = f(x)$$

$$y = \begin{cases} 1/2 & \in x = \{H, T\} \\ 0 & \in \text{otherwise} \end{cases}$$

Discrete RV.	Continuous RV.
Probability Mass function (PMF)	Probability Density function (PDF)

Cumulative probability

CMF
Cumulative Mass function

CDF
Cummulative Density function.

PMF :→ Probability Mass function

:→ It is a mathematical function that describe the probability distribution of a "Descrete Random Variable".

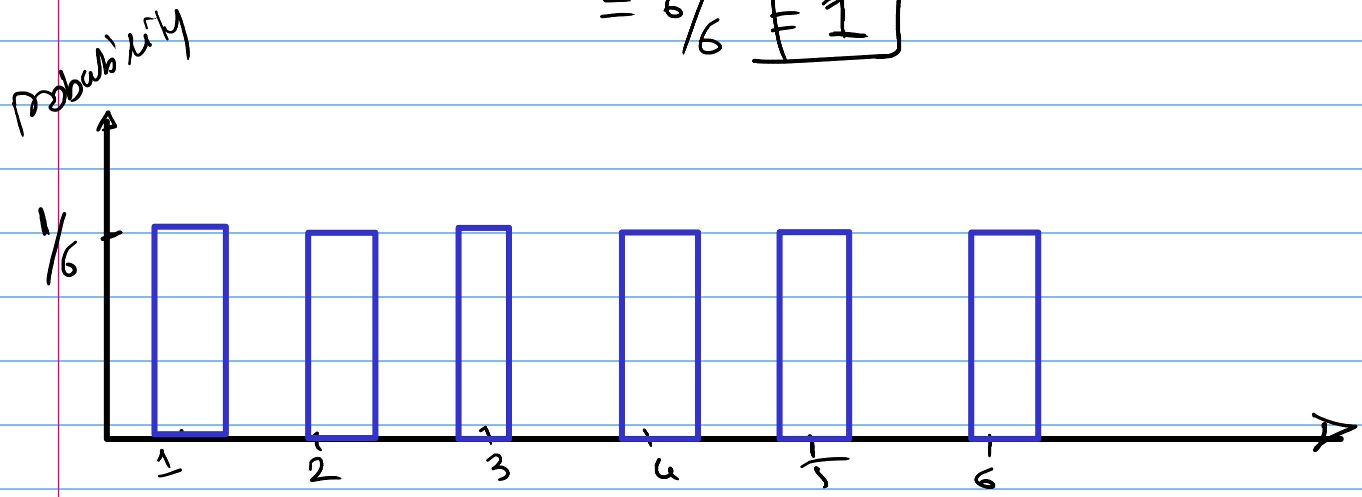
Condition 1 :→ The probability assign to each value must be non-negative.

:→ The sum of probability must be equal to 1.

Rolling a Dice $\{1, 2, 3, 4, 5, 6\}$

$P(1) = 1/6 \quad P(2) = 1/6 \quad P(3)\ 1/6$

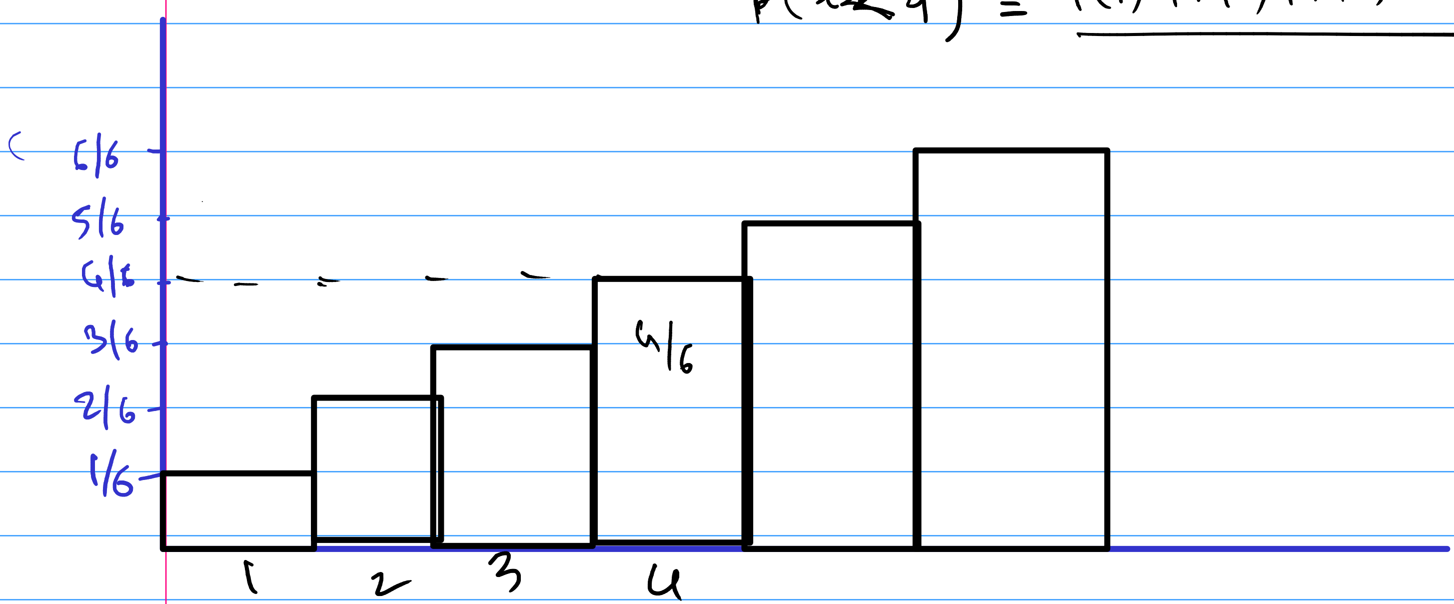
$$\text{Total prob} = P(1) + P(2) + P(3) + P(4) + P(5) + P(6)$$
$$= 1/6 + 1/6 + 1/6 + 1/6 + 1/6 + 1/6$$
$$= 6/6 = \boxed{1}$$



CMF :⇒

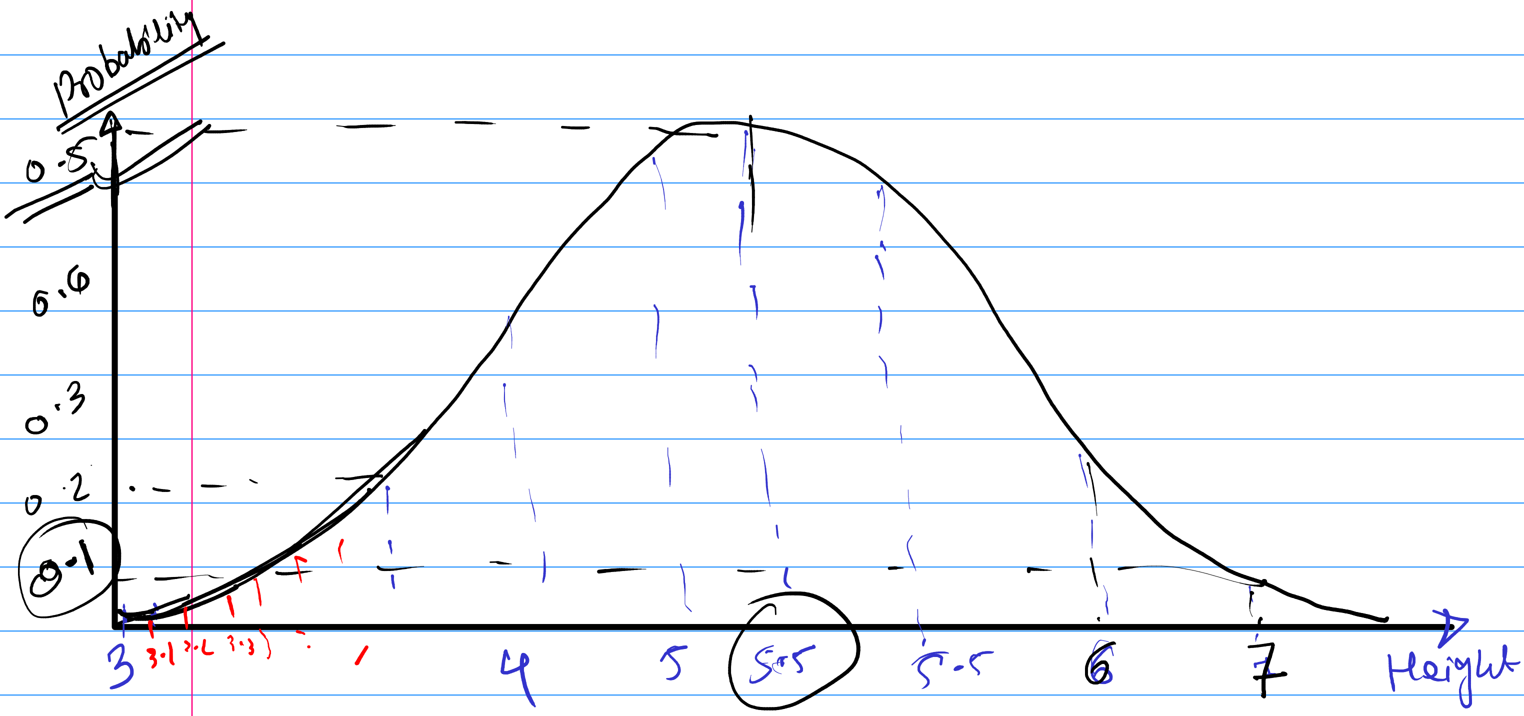
$$P(x \leq 4) = \underline{P(1) + P(2) + P(3) + P(4)}$$

6

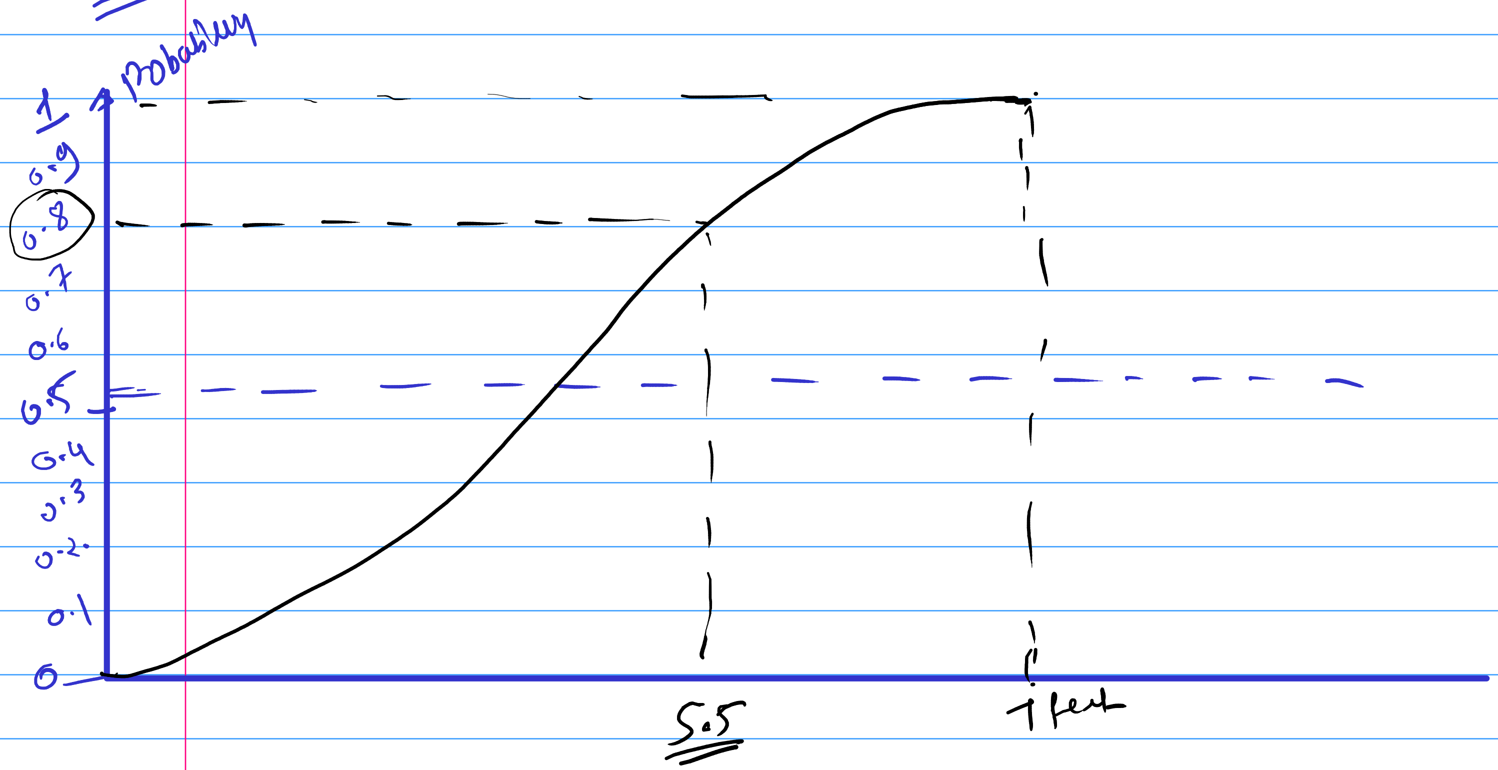


Probability Density function (PDF)

:→ It is a mathematical function that describe the probability distribution of a Continious Random Variable.



CDF → Cumulative Density function



Type of Distribution (probability Distribution)

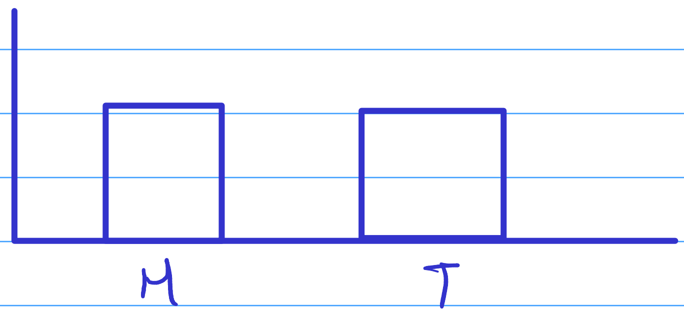
(1) Bernoulli Distribution :→

↳ It is a discrete probability Distribution

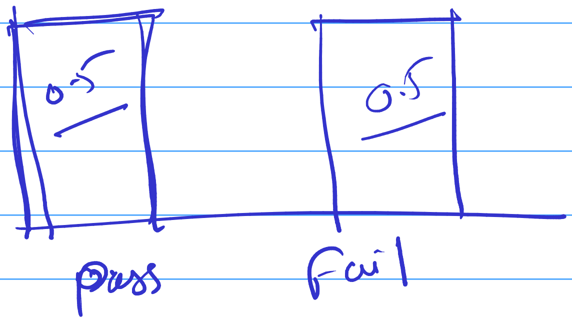
↳ It's Concerned with PMF.

↳ It applies to event that have one trial and two possible outcomes.

Coin → $P(H) = 0.5$
$P(T) = 0.5$



exam Result → Pass / fail

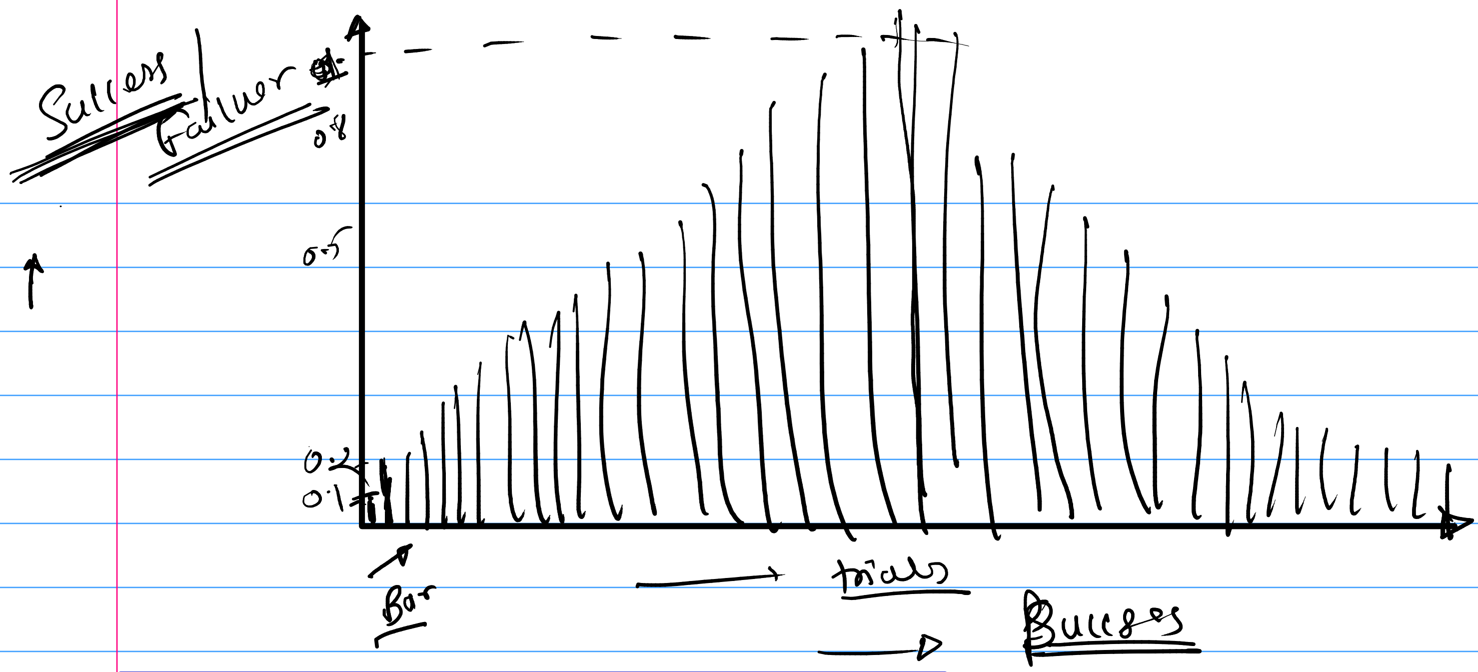


pass = 0.8
fail = 0.2

(2) Binomial Distribution :-

↳ It concerned with pmf.

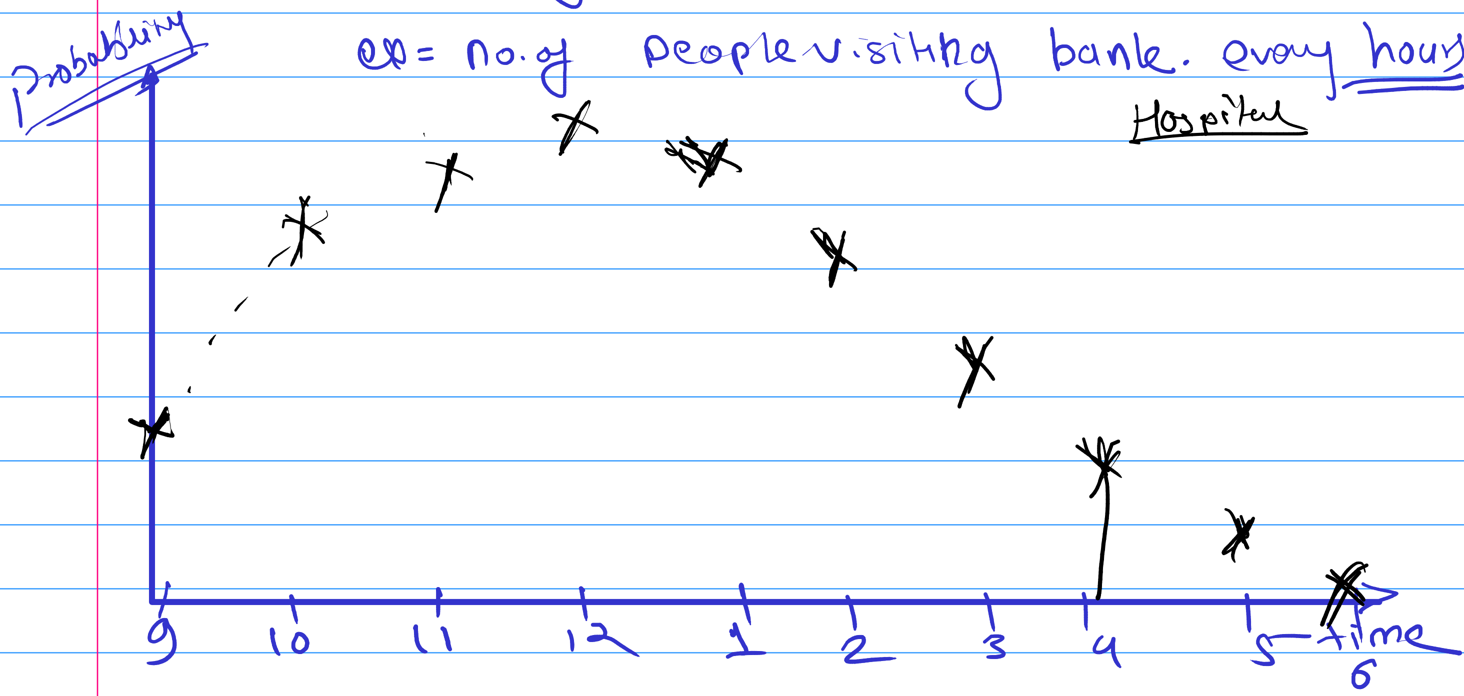
↳ It applies when we have two outcome but n. no. of trials (Independent trials)



③ Poisson Distribution

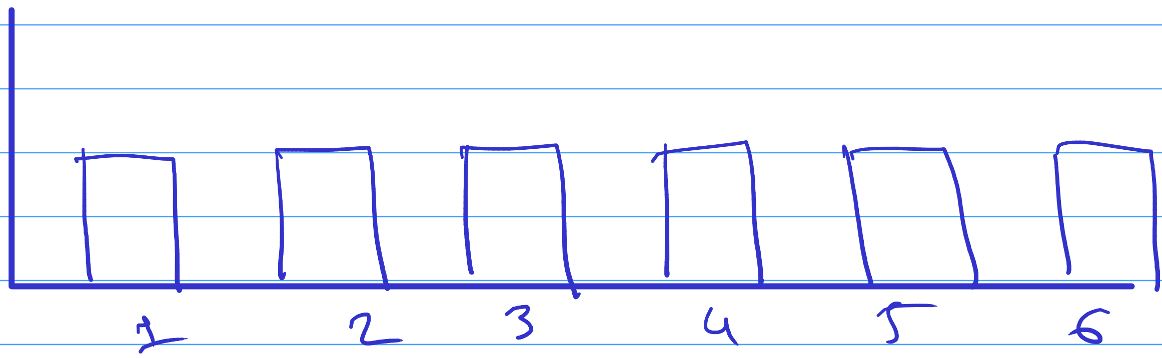
- It concerned with PMF
- It applies when the no. of events occuring in a fixed time interval.

eg = no. of people visiting bank. every hours
Hospital



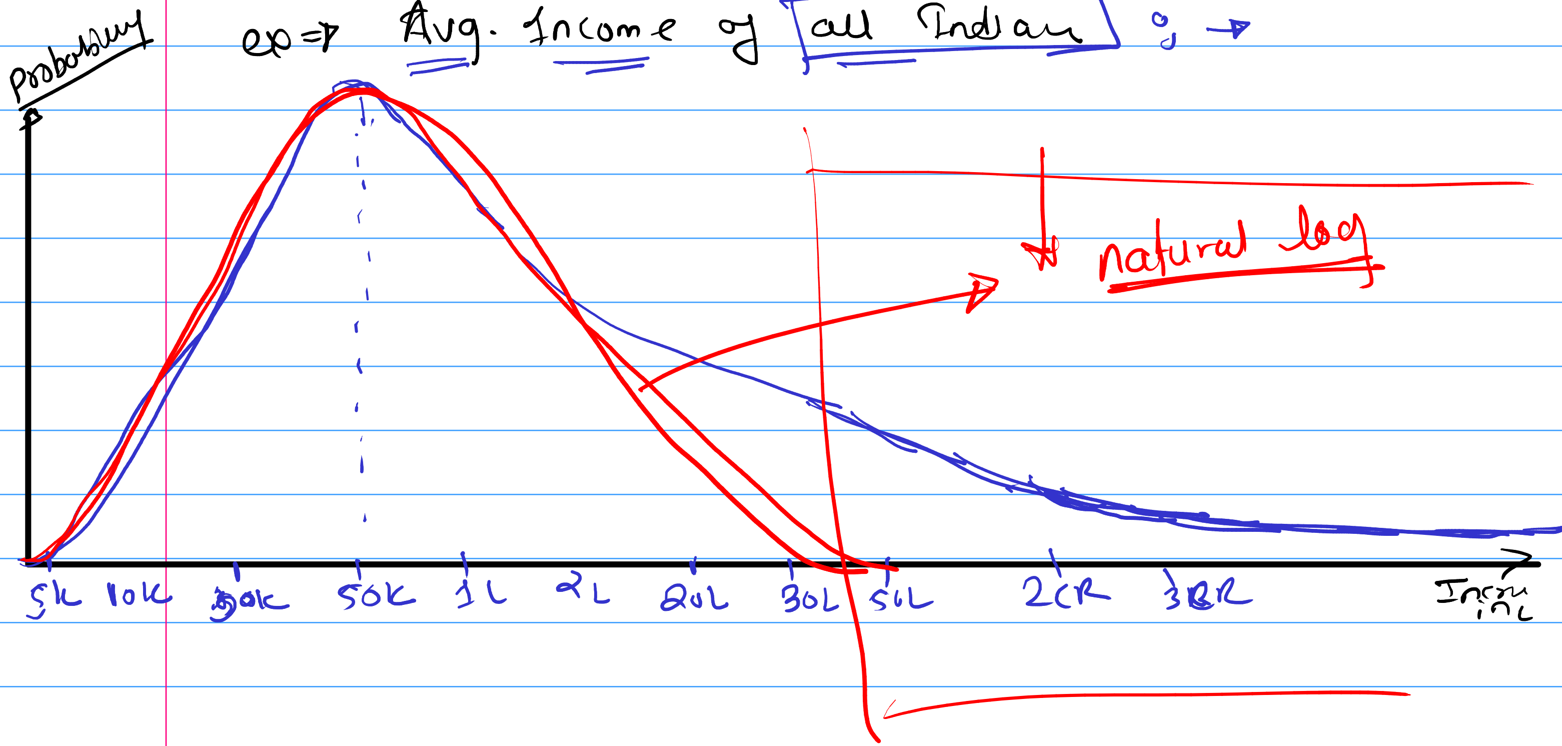
④ Uniform Distribution

Throwing a Dice (



⑤ Log Normal Distribution

ex ⇒ Avg. Income of all Indian : →



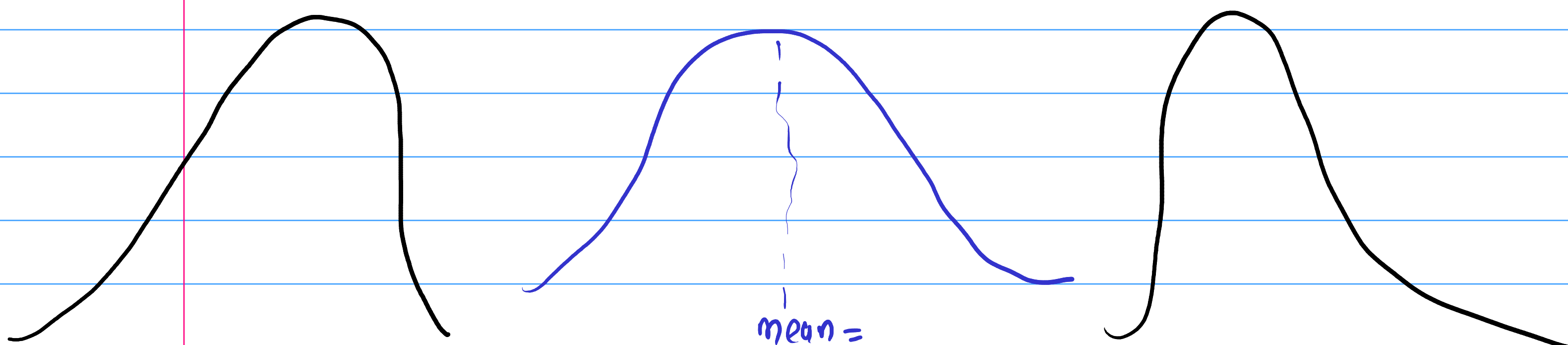
Normal / Guassian Distribution

↳ Continious RV (Height / weight / BP)

↳ It is Sysmmetrical Distribution from the Central tendency.

↳ Its look's like a <u>Bell curve</u>

$\boxed{\text{mean} = \text{median} = \text{mode}}$



mean < median < mode

mean = median = mode

mode < median < mean

<u>Left Skew</u>

<u>Right Skew</u>

<u>Skewness</u> → तिरछाप.

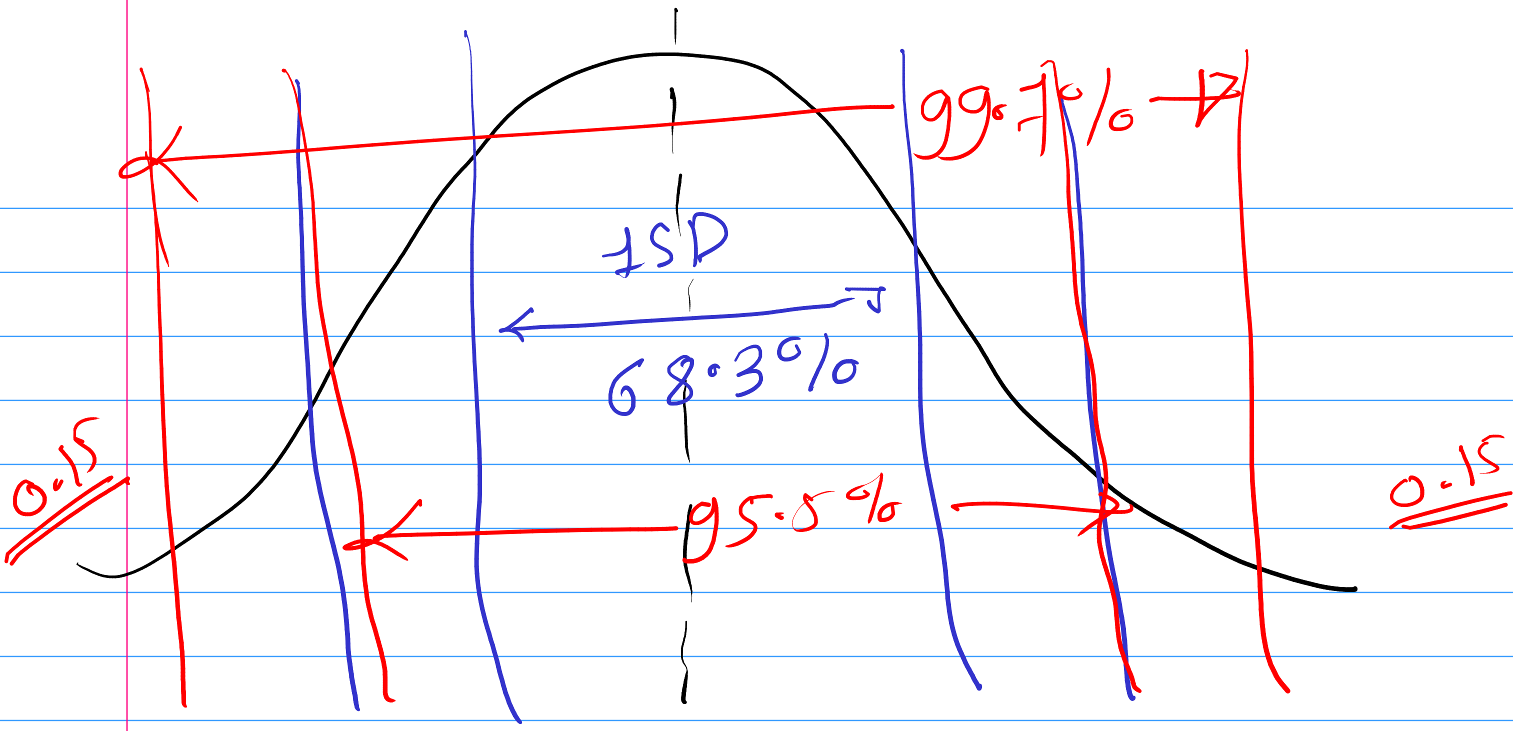
Empirical Rule of Normal Distribution

68.3, 95.5 & 99.7% Rule

The imperical rule States that for a normal distribution, 68.3% of the observation data will lie inside one standard deviation of mean,

95.5% lies inside Second SD.

& 99.7% lies within thirdd SD.



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